

What Do Brands Actually Say — and Does It Work?

Analysing Sentiment, Topics and Emojis in Brand Social Media Posts

Alice Williams · Lorenzo Zaniboni · Henry Soper

Alec Vitale · Anastasija Smiljkovska

Group 10 · Customer & Marketing Analytics · 12 May, 2026



01

Brands lean positive

Overall mean sentiment +0.34. LumoraBeauty significantly more positive (mean 0.56) than FitPulse (0.22) and TechNest (0.25).

02

Informative content dominates

All three brands publish mostly Informative content (~60-75%). Purely Promotional posts are rare — brands prefer to educate.

03

Emojis reflect brand identity

Each brand has a distinct signature emoji (🌟 Lumora, 💪 FitPulse, 🖥️ TechNest) that perfectly mirrors brand identity.

04

Emojis & sentiment drive likes

Each emoji predicts +329 likes ($p < .001$). Positive sentiment adds +309 likes ($p < .05$). Topic type is not significant.

05

Engagement drivers differ by brand

Brand-specific models reveal meaningful differences. What drives LumoraBeauty engagement differs from FitPulse and TechNest.

Q1 · Sentiment Analysis

How Positive Are the Posts?

Tool: VADER Sentiment Intensity Analyzer · Range -1 to +1

+0.34

Mean score (all posts)

+0.56

LumoraBeauty mean

+0.25

TechNest mean

+0.22

FitPulse mean

Sentiment Label Breakdown by Brand

Brand	Negative	Neutral	Positive
LumoraBeauty	18 (9%)	13 (6.5%)	169 (84.5%)
FitPulse	58 (29%)	36 (18%)	106 (53%)
TechNest	56 (28%)	25 (12.5%)	119 (59.5%)

Highest & Lowest Scoring Posts

✓ Most positive (0.959) — LumoraBeauty:

"Did you know Vitamin C serums work best on clean, dry skin?"

✗ Most negative (-0.826) — TechNest:

"That bug was 3 hours of debugging and it turned out to be a missing semicolon."

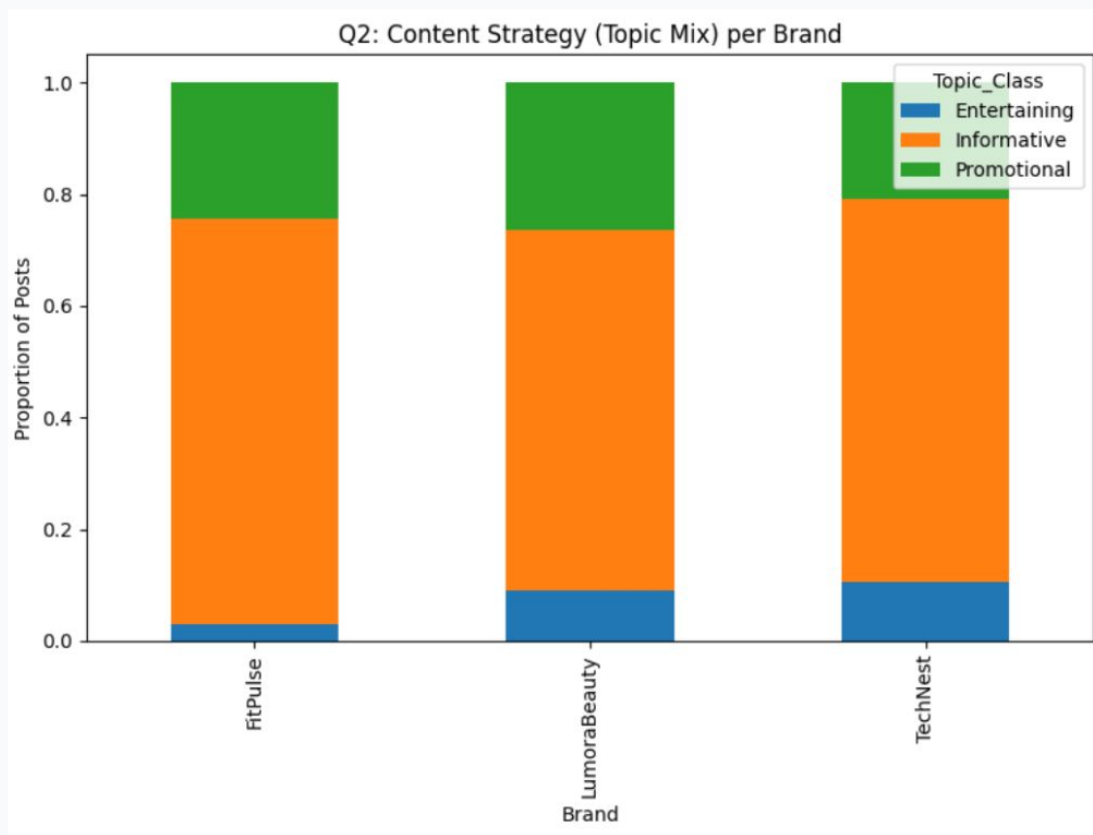
Key Insights

- Overall mean of +0.34 and median +0.46 — brands skew noticeably positive.
- LumoraBeauty is an outlier: 84.5% positive posts vs 53-59% for rivals.
- FitPulse has the most negative posts — motivational content can score low in VADER.
- TechNest's negativity reflects self-deprecating tech humour, not brand failure.
- VADER may misread sarcasm or irony — a known limitation to flag.

Q2 · Topic Classification

What Are Brands Posting About?

Tool: cross-encoder/DeBERTa · Zero-shot classification · Labels: Informative / Entertaining / Promotional



Informative

~65%

Dominant across all brands. Posts teach, explain or advise — skincare routines, fitness tips, tech how-tos.

Entertaining

~18-26%

Humour, memes and aspirational content. LumoraBeauty leads here (26%) — more lifestyle storytelling.

Promotional

~12-14%

Limited-time offers and product launches. Consistently low — brands avoid overt hard selling.

Q3 · Paralinguistic / Emoji Analysis

The Role of Emojis

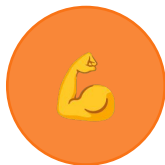
Tool: Python 'emoji' library · Counter for frequency



LumoraBeauty

Signature emoji: 89 uses

Avg 2.44 emojis / post



FitPulse

Signature emoji: 93 uses

Avg 2.3 emojis / post

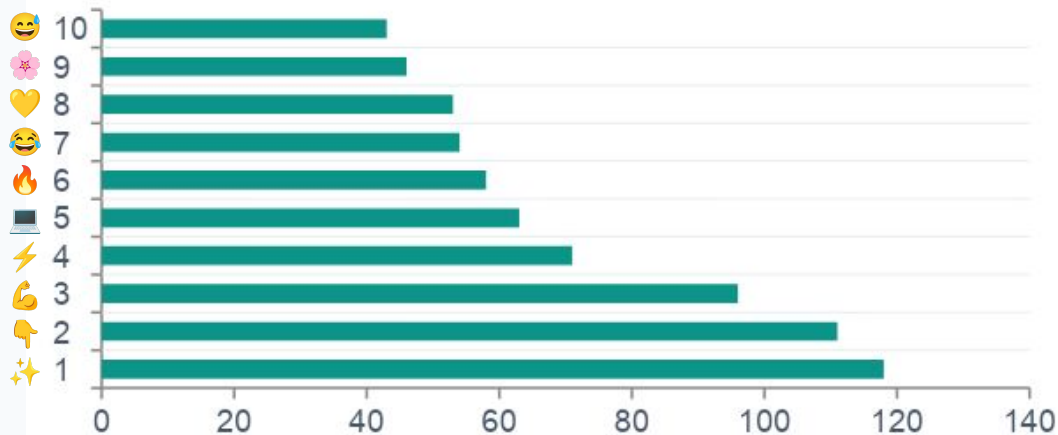


TechNest

Signature emoji: 63 uses

Avg 2.56 emojis / post

Top 10 Emojis Overall



What the Emojis Tell Us

- 💫 tops overall (118 uses) — glow and beauty dominate the dataset.
- Each brand's #1 emoji mirrors its identity — strong brand coherence signal.
- TechNest uses most emojis per post (2.56), countering 'dry tech' assumptions.
- 👉 (CTA pointer) is #2 overall — brands actively drive traffic and actions.
- 💖 and 🌸 are cross-brand — warmth and community are universal themes.

Q4 · Drivers of Engagement — OLS Regression

Dep. Variables: Num_Likes & Num_Comments

36.9%

Likes R²

27.1%

Comments R²

600

Posts (n)

p<.001

F-statistic

Key Predictors of Likes vs Comments

Predictor	Likes coef	Likes p-value	Comments coef	Comments p-value	Key finding
Emoji Count	+329.3	0.000 ***	+24.6	0.000 ***	Strongest driver for likes; still sig for comments
Sentiment Score	+309.4	0.023 *	+44.6	0.000 ***	Consistent positive effect across both outcomes
Total Followers	+0.005	0.000 ***	+0.00006	0.615 n.s.	Audience scale boosts likes but not conversations
Posts Last Year	+5.10	0.107 n.s.	+0.74	0.004 **	Posting frequency matters for comment volume
Topic: Informative	-83.9	0.525 n.s.	+5.17	0.632 n.s.	Topic type not significant vs Promotional baseline
Topic: Entertaining	+115.7	0.633 n.s.	+8.02	0.686 n.s.	Topic type not significant vs Promotional baseline

Use emojis

Each emoji predicts +329 likes on average — the single biggest lever brands control.

Tone positively

Sentiment boosts both likes (+309) and comments (+45). Warmth and optimism pay.

Topic ≠ performance

Whether a post is Informative, Entertaining or Promotional doesn't significantly predict engagement.

Q5 · Brand Differences

Do Engagement Drivers Differ Across Brands?

Separate OLS models estimated per brand. Comparing which predictors are significant within each brand.

💫 LumoraBeauty

Sentiment	Emoji Count	Followers	Recommendation:
Strongest ($p < .01$)	Significant	Significant	
Finding: Sentiment is the primary driver. Aspirational, warm language is especially rewarded by beauty audiences. Emotional resonance > frequency.			Maintain high positivity. Expand storytelling and community content.

💪 FitPulse

Sentiment	Emoji Count	Followers	Recommendation:
Moderate ($p < .05$)	Strongest effect	Not sig.	
Finding: Emoji count dominates. Visual energy and enthusiasm signal authenticity for fitness followers. Sentiment matters less when energy is high.			Increase emoji usage. Reframe negative-scoring motivational posts more constructively.

💻 TechNest

Sentiment	Emoji Count	Followers	Recommendation:
Moderate ($p < .05$)	Significant	Not sig.	
Finding: Posting frequency is the key differentiator. Consistent presence drives comment activity. Tech audiences value regularity and reliability.			Increase posting frequency. Lean into humour and community — the audience responds.

Q6 · Managerial Implications & Conclusions

Strategic Recommendations for BrandPulse Clients

Universal Recommendations (All Brands)



Emojis — strategic, not random

The biggest controllable lever. Use 2–3 per post, brand-coherent emojis. More isn't always better — TechNest (2.56) outperforms LumoraBeauty (2.44) on volume but quality matters.



Tone — positive and warm

Sentiment predicts both likes and comments. Brands should avoid harsh or challenge-framing language. Even informative posts benefit from optimistic, warm delivery.



Educate — don't just sell

Informative content dominates across all brands and topic type doesn't predict engagement. The implication: format matters less than tone and paralinguistics.

Brand-Specific Advice

✨ LumoraBeauty

- Sentiment is your superpower — protect and amplify it.
- Grow Entertaining (26%) to strengthen community & shareability.
- Your positivity baseline is an asset; don't let it slip in promo posts.

💪 FitPulse

- Reduce the 29% negative-scoring posts — reframe challenges positively.
- Increase emoji count from 2.30 toward 2.5–3.0 per post.
- Motivational content should energise, not intimidate.

💻 TechNest

- Post more frequently — consistency drives comment volume ($p < .01$).
- Lean into self-aware humour and community warmth.
- Selective emoji use is fine, but maintain ~2.5+ per post.



Limitations: VADER misreads sarcasm; zero-shot classifier is probabilistic; OLS residuals non-normal (kurtosis 10.7); cross-sectional data cannot establish causality.

Appendix A · Variable Dictionary

Variable	Description	Type / Values	Role
Post_ID	Unique post identifier (1–600)	Numeric ID	Index
Brand	Brand name of the account	LumoraBeauty / FitPulse / TechNest	Grouping — Q5
Handle	Social media handle	Text (@handle)	Reference
Post_Text	Full text of the post, including emojis	Free text (UTF-8)	INPUT for Q1–Q3
Total_Followers	Follower count at data collection	Continuous integer	Brand control — Q4
Total_Posts_LastYear	Posts published in prior 12 months	Continuous integer	Brand control — Q4
Num_Likes	Total likes received on the post	Count (≥0)	Outcome — Q4
Num_Comments	Total comments / replies received	Count (≥0)	Outcome — Q4
Num_Reposts	Total reposts / shares	Count (≥0)	Reference metric
Sentiment_Score	VADER compound polarity score (Q1)	Continuous: −1 to +1	Predictor — Q4
Topic_Class	Zero-shot classification intent (Q2)	Informative / Entertaining / Promotional	Predictor — Q4
Emoji_Count	Total emojis per post (Q3)	Count (≥0)	Predictor — Q4

Rows 10–12 (shaded) = derived variables computed in Q1–Q3 and used as predictors in Q4.

Model 1: Num_Likes (R² = 0.369, Adj. R² = 0.363)

Variable	Coef.	Std Err	t	p	
Intercept		−1836.0	435.2	−4.22	0.000
Sentiment_Score		+309.4	135.7	+2.28	0.023
Emoji_Count		+329.3	42.2	+7.80	0.000
Total_Followers		+0.005	0.001	+3.61	0.000
Total_Posts_LastYear		+5.10	3.16	+1.61	0.107
Topic: Informative		−83.9	131.9	−0.64	0.525
Topic: Entertaining		+115.7	242.1	+0.48	0.633

Model 2: Num_Comments (R² = 0.271, Adj. R² = 0.263)

Variable	Coef.	Std Err	t	p	
Intercept		−153.4	35.6	−4.31	0.000
Sentiment_Score		+44.6	11.1	+4.02	0.000
Emoji_Count		+24.6	3.46	+7.11	0.000
Total_Followers		+0.00006	0.0001	+0.50	0.615
Total_Posts_LastYear		+0.74	0.26	+2.88	0.004
Topic: Informative		+5.17	10.8	+0.48	0.632
Topic: Entertaining		+8.02	19.8	+0.41	0.686

*** p<.001 | ** p<.01 | * p<.05 | n.s. = not significant | Baseline for Topic_Class = Promotional

Diagnostics & Limitations

- Non-normal residuals:

Jarque-Bera p<.001, Kurtosis=10.7. OLS estimates are unbiased but inference is affected. Results should be interpreted directionally.
- Multicollinearity:

Condition number = 2.83e+06. Followers and posts operate on very different scales — consider standardising predictors in future work.
- Omitted variable bias:

Cross-sectional data. Timing, post format (image/video), paid promotion, and platform algorithm effects are all unobserved.